Michael W. Lu

Full-stack engineer and technical lead with 5+ years building federal systems at USDA and NSF | Active Secret clearance

San Francisco, CA | (214) 708-0771 | michaelwenlu@gmail.com | michaelw.lu | github.com/michaelwlu | linkedin.com/in/michaelwlu

Experience

Spatial Front, Inc. | Software consultancy serving federal agencies | Remote | Feb 2021 – present

Software Engineer (2021) → Front-End Technical Lead (2021) → Technical Product Manager (2022, current title)

Role: **Technical Lead & Deputy Project Manager** | Client: National Science Foundation | Jul 2025 – present

- Direct a 10-person SharePoint development and administration team serving 3,000 staff and 8,900 external guests
- Led Microsoft Copilot readiness for 6,500 tenant sites: ranked them by PII risk and content volume, temporarily restricted Copilot on the 570 riskiest sites (87% of content), and launched phased permission reviews based on priority
- Designed a site provisioning workflow that replaced manual setup with ServiceNow intake, one-click approval in a SharePoint list, and automated site creation in Power Automate, used for 150+ sites to date
- Evaluated Okta MFA integration for 3,000 fellowship reviewers with security and infrastructure teams, found it would split guests unpredictably across two sign-in paths, and persuaded federal leads to standardize all 8,900 guests on Entra ID

Role: **Tech Lead Manager** | Client: U.S. Department of Agriculture | Feb 2021 – Jul 2025

- Managed and mentored 20 people across two cross-functional teams and directed architecture for systems with 15,000 monthly users in 2,500 field offices, including overseeing a .NET Framework to .NET 6 migration
- Stepped in to turn around a stalled rebuild of a QuarkXPress-based document publishing app on .NET, React, and SharePoint, replacing the legacy system within 11 months and eliminating $300,000 in annual licensing costs
- Architected a relational schema powering a workflow engine for 60,000 documents across 16 workflows and 53 types
- Built Elasticsearch full-text search across 160,000 documents, making document text searchable alongside metadata
- Implemented batch create, move, and archive for up to 25 documents at once, replacing one-at-a-time processing
- Modernized the legacy React codebase with hooks, paying down major technical debt and cutting component size by 78%

Programming Sabbatical | New York, NY | Jan 2020 – Feb 2021

- Learned web development through self-study and pair programming at the Recurse Center, a 12-week programming retreat
- Built full-stack apps, including Definitely Not Reddit (Next.js, GraphQL, PostgreSQL) and a small-business website

NYU Langone Health | New York, NY | Mar 2014 – Jan 2020

Financial Analyst (2014) → Senior Financial Analyst (2017)

- Developed a forecasting system in Excel, with one master workbook refreshing dashboards for 32 research labs, replacing manual reporting with two-year projections so faculty could catch funding gaps a full grant cycle early
- Managed research grant finances for the Neuroscience Institute as funding doubled from $20M to $40M in 6 years

Skills

Languages: JavaScript, TypeScript, C#, Python, SQL, HTML/CSS

Front End: React, Next.js, TanStack Query, Tailwind CSS, Jest, React Testing Library, Section 508/WCAG

Back End & Data: ASP.NET Core Web API, Entity Framework, xUnit, SQL Server, PostgreSQL, Elasticsearch

Tools & Platforms: AWS (EC2, S3), SharePoint Online, Git, Jenkins, Jira, Agile/Scrum, GitHub Copilot, Claude Code

Education

New York University, Stern School of Business | B.S. in Finance & Economics, 2013

Certifications & Awards: SAFe 5 Practitioner | Spatial Front hackathons: 1st place ×2, 2nd place ×5 (2021 – 2023)

Community

Code Nation | Volunteer Instructor | New York, NY | Oct 2022 – May 2024

- Taught web development to under-represented NYC students, with 67% of program alumni entering STEM vs. 8% baseline